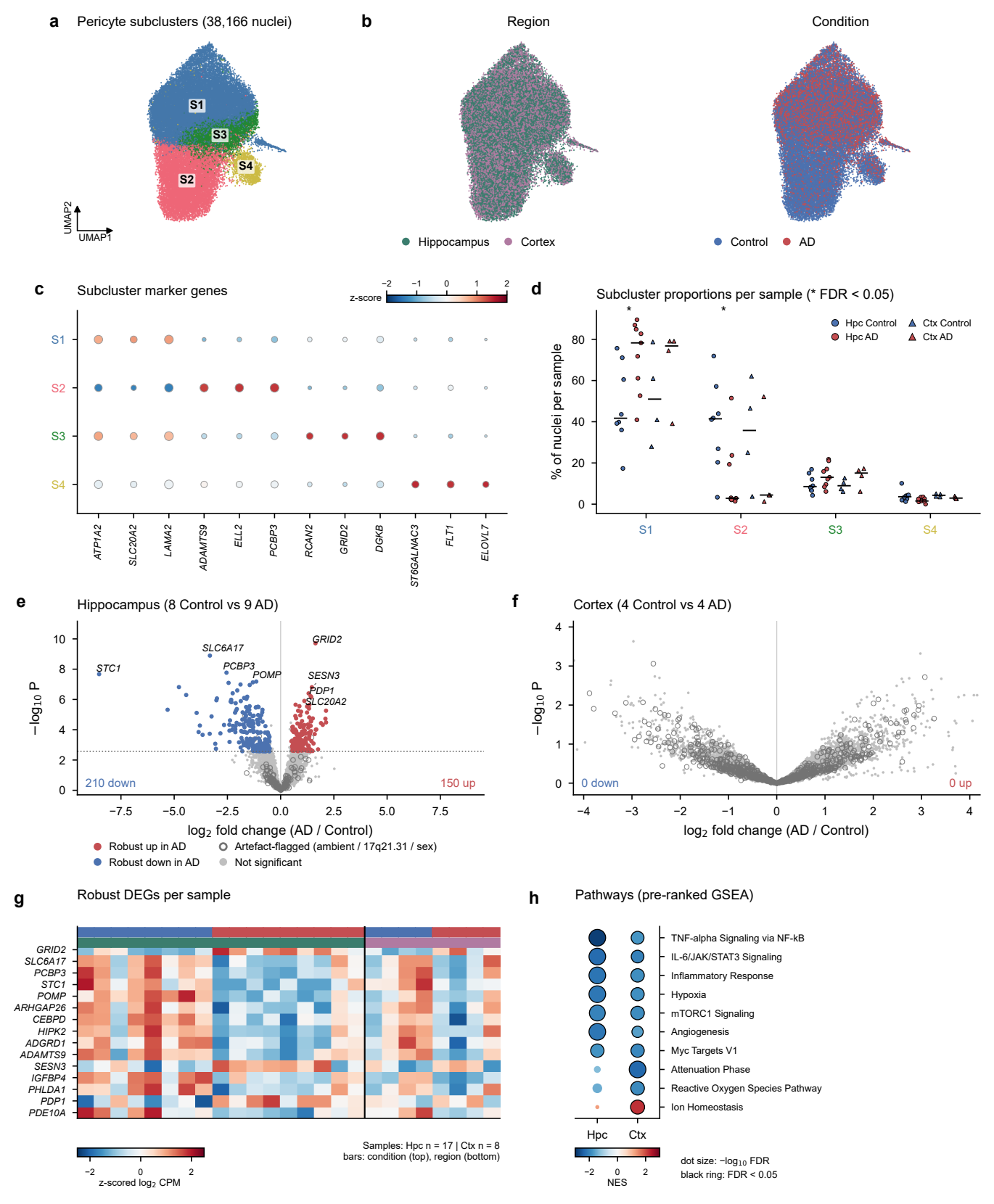

a
Pericyte subclusters (38,166 nuclei)
S1
S3
S4
S2
UMAP2
UMAP1
b
Region
Hippocampus
Cortex
Condition
Control
AD
c
Subcluster marker genes
z-score
−2 −1 0 1 2
S1
S2
S3
S4
ATP1A2
SLC20A2
LAMA2
ADAMTS9
ELL2
PCBP3
RCAN2
GRID2
DGKB
ST6GALNAC3
FLT1
ELOVL7
d
Subcluster proportions per sample (* FDR < 0.05)
Hpc Control
Hpc AD
Ctx Control
Ctx AD
% of nuclei per sample
0 20 40 60 80
S1
S2
S3
S4
e
Hippocampus (8 Control vs 9 AD)
STC1
SLC6A17
PCBP3
POMP
GRID2
SESN3
PDP1
SLC20A2
210 down
150 up
−log10 P
log2 fold change (AD / Control)
Robust up in AD
Robust down in AD
Artefact-flagged (ambient / 17q21.31 / sex)
Not significant
f
Cortex (4 Control vs 4 AD)
0 down
0 up
−log10 P
log2 fold change (AD / Control)
g
Robust DEGs per sample
GRID2
SLC6A17
PCBP3
STC1
POMP
ARHGAP26
CEBPD
HIPK2
ADGRD1
ADAMTS9
SESN3
IGFBP4
PHLDA1
PDP1
PDE10A
z-scored log2 CPM
Samples: Hpc n = 17 | Ctx n = 8
bars: condition (top), region (bottom)
h
Pathways (pre-ranked GSEA)
TNF-alpha Signaling via NF-kB
IL-6/JAK/STAT3 Signaling
Inflammatory Response
Hypoxia
mTORC1 Signaling
Angiogenesis
Myc Targets V1
Attenuation Phase
Reactive Oxygen Species Pathway
Ion Homeostasis
Hpc
Ctx
NES
dot size: −log10 FDR
black ring: FDR < 0.05